Raghav Chhabra

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EDUCATION

Thapar Institute of Engineering and Technology Patiala, India
Bachelor of Engineering in Computer Engineering — CGPA: 8.89/10 *Aug 2023 – May 2027*

TECHNICAL SKILLS

Programming & Query Languages: Python (Advanced), SQL, JavaScript, C, C++, HTML/CSS
Data & ML Libraries: Pandas, NumPy, Scikit-learn, PyTorch, OpenCV, Matplotlib, Seaborn
Data Science Skills: EDA, Feature Engineering, Statistical Analysis, Model Evaluation, Data Visualization, Pipeline Automation
Frameworks & Platforms: Flask, Streamlit
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

WORK EXPERIENCE

Axis Bank – Business Intelligence Unit May 2026 - Ongoing
Data Science Intern
- Upcoming role focused on predictive analytics, hyper-personalization, and customer behavior modeling - analyzing large-scale banking data using Python and SQL to support strategic planning.

OutriX Jul 2025 – Aug 2025
AI/ML Intern (Remote)
- Designed and trained a custom CNN model using PyTorch for CIFAR-10 image classification across 10 categories, building a full end-to-end pipeline from raw data ingestion to evaluated output.
- Automated dataset preprocessing and training workflows using optimized data loaders, cutting manual steps and improving pipeline efficiency.
- Documented evaluation metrics, model performance, and experimental insights through structured weekly reports, translating technical findings into clear business summaries for stakeholders.

PROJECTS

Energy Consumption Forecasting using Machine Learning UCI Household Electric Power Consumption Dataset
- Built an end-to-end forecasting pipeline on millions of time-series records; engineered 80+ features including lag variables, rolling statistics, EWMA, and cyclical encoding to capture complex demand patterns.
- Evaluated seven regression models and achieved 99.53% forecasting accuracy using Ridge Regression, providing insights directly applicable to demand forecasting and strategic resource allocation.
- Automated the entire workflow from raw data ingestion to model evaluation, creating a repeatable, production-ready pipeline that eliminates manual data tasks.

Book Recommendation & Customer Behavior Analysis Goodbooks-10k Dataset
- Built a personalized recommendation engine using TF-IDF vectorization, cosine similarity, and SVD-based collaborative filtering (RMSE: 0.84) to model user preferences and predict future behavior.
- Applied NLP techniques to unstructured review text to identify sentiment patterns and preference signals for diagnostic insight into customer reviews.
- Used graph-based modeling with NetworkX and A* search to uncover underlying user choice patterns, translating complex network structures into actionable analytical insights.

ACHIEVEMENTS

- Selected among the Top 450 students from 20,000+ applicants for the AlgoUniversity Technology Fellowship.
- National Semi-Finalist in Flipkart GRID 7.0 from over 160,000 participants.
- Ranked within the Top 15,000 among 5M+ users on LeetCode with 1,000+ solved problems - demonstrating strong algorithmic thinking and problem-solving efficiency.
- Selected for the McKinsey Forward Program by McKinsey & Company - recognizing analytical thinking, business problem-solving aptitude, and leadership potential.

RELEVANT COURSEWORK

Artificial Intelligence | Machine Learning | Data Structures and Algorithms | Object-Oriented Programming | Database Management Systems | C/C++ Programming